Hierarchical Heterogeneous Geometric Foreground Perception Network for Remote Sensing Object Detection

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***Abstract*— Recently, deep learning-based remote sensing object detection (RSOD) has been widely explored and obtained remarkable performance. However, most existing multiscale feature extraction methods neglect exploring the interfering representation of different hierarchical features in the backbone, which is crucial for learning more discriminative features. Moreover, feature pyramid network (FPN) and its variants have difficulty in effectively perceiving the pose and salient information of remote sensing objects, leading to reduced detection accuracy. To address these issues, we propose a hierarchical heterogeneous geometric foreground perception network (HHGFP-Net) for RSOD. Specifically, a hierarchical heterogeneous receptive-field module (HHRM) is proposed to reward and penalize the feature information of the corresponding levels according to the differences between the shallow and deep feature layers in the backbone, improving discriminative feature ability. Furthermore, a geometric foreground perception FPN (GFP-FPN) is developed to refine geometric shapes and enhance foreground contents, providing more precise feature representations for objects, particularly small objects. Experimental results on four challenging RSOD datasets demonstrate that our HHGFP-Net achieves state-of-the-art performance. Codes are available at: https://github.com/YyLinkWorld/HHGFP-Net.**

***Index Terms*— Foreground enhancing, geometric feature perception, hierarchical heterogeneous, multiscale feature extraction, remote sensing object detection (RSOD).**

I. Introduction

REMOTE sensing object detection (RSOD) is dedicated to discerning the presence of specific objects from aerial or satellite imagery, and subsequently ascertaining their category and location. Due to its significant value in practical applications such as military reconnaissance [1], urban planning [2], and traffic guidance [3], RSOD has long received extensive attention from researchers. However, owing to the problems of large scale variations, arbitrary orientations, and uneven distribution density of objects, accurately recognizing objects in aerial images remains a challenging topic.

Early works in RSOD focused on adopting generic frameworks like YOLO [4] and R-CNN [5] to detect objects in remote sensing images (RSIs). Nevertheless, these detectors [4], [5], [6], [7], [8], [9] designed for natural images are not suitable to processing RSIs due to differences in application areas. Specifically, traditional object detection frameworks cannot accurately describe objects in arbitrary orientations [10], nor can they handle the significant variations in object scales [11], [12]. Therefore, recent research trends have shifted toward developing specialized detectors for RSIs, including designing feature extraction networks appropriate for RSOD and improving oriented bounding box (OBB) representations [11], [13], [14], [15], [16]. Despite achieving good results, there are still two unresolved issues that demand further exploration.

1) *Prevailing Multiscale Feature Extraction Methods Exhibit Feature Representation Confusion in Different Layers of the Backbone, Leading to Insufficient Feature Discriminative Capability:* In the past decade, general visual backbones like ResNet [17] and VGG [18] have been widely used in RSOD. However, these methods share the same receptive field for objects of varying sizes, which limits their effectiveness in handling object scale variations in RSIs. To address this, recent advanced RSOD [11], [13], [19], [20], [21] advocate integrating multiscale feature extraction into the backbone to learn features of objects at different scales, resulting in notable improvements. Nevertheless, these methods overlook a critical issue: features of different scales can exist concurrently at any level of the backbone and cause mutual interference. Specifically, large kernels in shallow layers introduce global features while marginalizing local details, which is detrimental to detecting small objects. In deeper layers, local features with small receptive fields can interfere with overall perception, leading to poor performance in detecting large objects. Therefore, further research is needed to clarify appropriate feature representations and to reduce interference during multiscale feature extraction.

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2) *How to Design an FPN That Effectively Enhances Object Pose and Prominent Foreground Perception, Achieving More Accurate Object Feature Representation?:* Due to its effectiveness in bridging the semantic gap between features at different levels, the FPN [22] structure is widely used in downstream tasks such as object detection and semantic segmentation. However, the model design and hyperparameter settings of existing FPNs [22], [23], [24], [25], [26] are typically learned from natural images, limiting their potential in RSIs. Specifically, since RSIs are captured from an aerial perspective, objects often exhibit various shapes and arbitrary orientations. Previous methods [10], [15], [27] addressing this issue typically focus on improving loss functions and bounding box regression during postprocessing, while lacking effective learning, these characteristics before final detection. Furthermore, in RSIs, small object size and background occlusion can lead to insufficient representation of salient features during detection, which is crucial for accurately identifying objects, especially small objects. As a result, effectively learning the geometric features of objects and enhancing foreground feature representations before final detection remains an unresolved issue.

To address the above mentioned issues, in this article, we propose a hierarchical heterogeneous geometric foreground perception network (HHGFP-Net) for RSOD. Specifically, a hierarchical heterogeneous receptive-field module (HHRM) is proposed, which operates with different strategies across network layers. In the shallow layers of the backbone, the module leverages hierarchical information to apply a positive offset to the weights of smaller receptive fields, enhancing local features, while introducing a negative offset to the weights of larger receptive fields to suppress global features. In contrast, the strategy is reversed in the deeper layers. Moreover, we propose a GFP feature pyramid network (GFP-FPN), which models the geometric shapes of remote sensing objects with dynamic and irregular feature encoders, and it then searches for consistent information across multilevel semantics to enhance prominent foreground features.

The main contributions of this work are summarized as follows.

1) With the well-designed HHRM, we leverage hierarchical information to apply weight offsets across different feature layers, enhancing critical features and suppressing interference, making the features more discriminative.
2) A GFP-FPN is proposed to fit the geometric pose of objects in RSIs and enhance foreground feature representation by searching for consistency information across multiple feature levels, improving detection accuracy of objects, especially for small objects.
3) We propose a novel RSOD framework called HHGFP-Net, which seamlessly integrates a well-designed HHRM and GFP-FPN. Experiments on challenging datasets, including DOTA-v1.0 [12], DOTA-v1.5 [12], DIOR-R [28], and STAR [29], demonstrated that our method achieves superior performance compared to several state-of-the-art methods.

II. Related Work

In this section, we briefly review some related works on generic and RSOD, including those dedicated to improving feature extraction and necking FPNs to enhance fitness to RSIs.

A. Generic Object Detection

Thanks to the advancements in deep learning, object detection has witnessed significant development in recent years. Most existing generic object detectors are primarily focused on natural image detection and can be roughly categorized into two types: two-stage [5], [30] and single-stage detectors [4], [7], [8], [9], [31]. Two-stage methods, such as Faster RCNN [32], first generate candidate boxes for objects using a region proposal network (RPN) and then classify objects within those boxes. Generally, two-stage methods achieve higher detection accuracy but are computationally complex and not suitable for real-time inference [10]. To address the high inference cost, single-stage detectors like YOLO [4] and SSD [8] have been proposed. These methods estimate object positions and perform classification simultaneously, significantly improving speed but sacrificing some accuracy. The recent RT-DETR [33] achieves fast processing of multiscale features by decoupling intrascale interactions and integrating cross-scale information, effectively enhancing speed without compromising accuracy. However, the aforementioned methods are all anchor-based, and the preset anchors may not effectively match the diverse objects in images. Therefore, some anchor-free detection methods have been proposed, such as FCOS [34] and CornerNet [35], which define anchor prediction as a regression to the object's center point and the width and height around it, avoiding the missed detections or redundant detections caused by unreasonable anchor settings.

Due to the arbitrary orientations and shapes of objects in RSIs, horizontal bounding boxes (HBB) designed for natural images fail to accurately capture these characteristics. As a result, recent mainstream approaches in RSOD have shifted toward developing specialized OBB detection frameworks, achieving remarkable progress.

B. Remote Sensing Object Detection

Due to differences in application scenarios and imaging methods, object detectors trained on natural images are unsuitable for RSIs. Therefore, some detection methods targeting OBBs have been proposed [10], [15], [16], [27], [36], [37], focusing on aligning features with the object's orientation during postprocessing and achieving significant progress. More recently, following an in-depth study of the intrinsic characteristics of RSIs, outstanding success has been achieved in the design of feature extraction backbones for RSOD [9], [11], [13], [14], [38], [39], [40], [41], [42]. Most influential research focuses on addressing significant scale variation in objects within RSIs, emphasizing the importance

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of large receptive fields and multiscale feature extraction for RSOD. For instance, LSKNet [13] incorporates large convolutional kernels into its backbone to capture extensive contextual information, aiming to accurately identify targets through their surrounding environment. PKINet [11] integrates multiscale feature perception into its backbone for modeling objects of various scales and has achieved the remarkable performance to date.

Although multiscale feature extraction combined with large kernels can effectively address the traditional framework's insufficient response to large-scale object variation in RSIs [11], it also leads to the coexistence and mutual interference of global and local features at different backbone levels. Unlike previous methods, we propose HHRM, which can selectively enhance important features at different backbone levels and suppress interference representations under the guidance of hierarchical information, thereby avoiding these issues.

C. Feature Pyramid Network

The FPN [22]) is a deep learning model widely used in object detection, semantic segmentation, and other downstream tasks, efficiently integrating features from different scales to enhance algorithm performance. To further improve the complementarity of information between features at different scales, many outstanding variants of FPN (PANet [43], NAS-FPN [24], BiFPN [23]) have been proposed. For instance, the AFPN [25] progressively integrates features from adjacent scales, mitigating semantic disparities during feature fusion. AugFPN [26] employs a single-shot supervision approach to address the misalignment of local details with high-level semantics. In the field of RSOD, some researchers have also explored the applicability of the generic FPN to RSIs. DN-FPN [44] effectively reduces the impact of noise on feature representation through geometric semantic contrastive learning, improving the detection accuracy of small targets. GCWNet [45] repeatedly fuses features of different orders in the neck and refines them to enhance representation. Despite these advancements, research on optimizing FPN to better meet the feature needs of RSIs remains limited.

To enhance the applicability of general FPNs in RSOD, this study proposes a GFP-FPN. It focuses on object pose fitting and significant foreground feature enhancement to provide a more accurate object representation.

D. Geometric Pose and Shape Learning

Objects often exhibit arbitrary orientations and poses in RSIs, while background noise frequently interferes with the detection results. To address these challenges, recent research has focused on improving the adaptability of detection models to geometric shapes and mitigating the impact of background noise [27], [46], [47], [48], [49], [50]. In postprocess optimization, many methods refine the detection accuracy by adjusting the bounding box regression strategies and label assignment mechanisms. EGPNet [51] employs a progressive scale-feature fusion encoder for feature extraction and introduces an edge-guided image refinement module, enhancing object shape completeness while suppressing background interference. DCFL [15] introduces a dynamic prior capture and label assignment strategy, effectively capturing complex object geometries. RSDet [52] alleviates rotation-induced fitting errors by employing an improved bounding box regression loss (modulated loss), enabling more accurate pose and orientation descriptions. GWD [49] tackles discontinuities in bounding box regression, while KLD [53] dynamically adjusts gradient distributions based on object characteristics to improve fitting precision. In geometric modeling, some approaches enhance the representation of object geometry prior to final detection. For instance, GCWNet [45] integrates multiple branches into the neck of the detection network to strengthen shape perception through contextual modeling. AO2-DETR [54] leverages the Transformer architecture, employing dynamic feature sampling to model object shapes efficiently before detection.

III. Method

In this section, we first introduce the overall architecture of the HHGFP-Net in Section III-A, then elaborate on the proposed HHRM in Section III-B, detailed the GFP-FPN in Section III-C, and finally discuss the applied loss functions in Section III-D.

A. Overall Framework

In this article, we construct HHGFP-Net using O-RCNN [36] as the baseline. As shown in Fig. 1, our proposed HHGFP-Net is an end-to-end RSOD framework that takes RGB images as input, and draws OBBs around the objects of interest in the output image.

In HHGFP-Net, we propose the HHRM (see Fig. 2) as the foundational component of the feature extraction network. HHRM employs branches with varying receptive fields and incorporates independent positional encodings to address feature confusion arising from mutual interference between different scales. By integrating the hierarchical reward and penalty mechanism (HRPM), HRPM adaptively enhances essential local features in shallow layers to improve the detection of small objects, while suppressing global information, and, conversely, amplifies global features in deeper layers to facilitate large object detection, simultaneously attenuating local feature representations. To further advance geometric pose perception, which has been underexplored in predetection stages, we introduce the geometric feature refinement block (GFRB) within GFP-FPN (see Fig. 3). The GFRB enables the early modeling of spatial and geometric properties of objects, thereby optimizing feature representations before the detection head and reducing the reliance on complex postprocessing. Complementing this, we develop the prominent foreground enhancement block (PFEB), which aligns consistent features across multiple semantic levels and highlights salient foreground information. Together, these modules provide a more precise and robust multiscale feature representation, significantly improving the accuracy and reliability of object detection.

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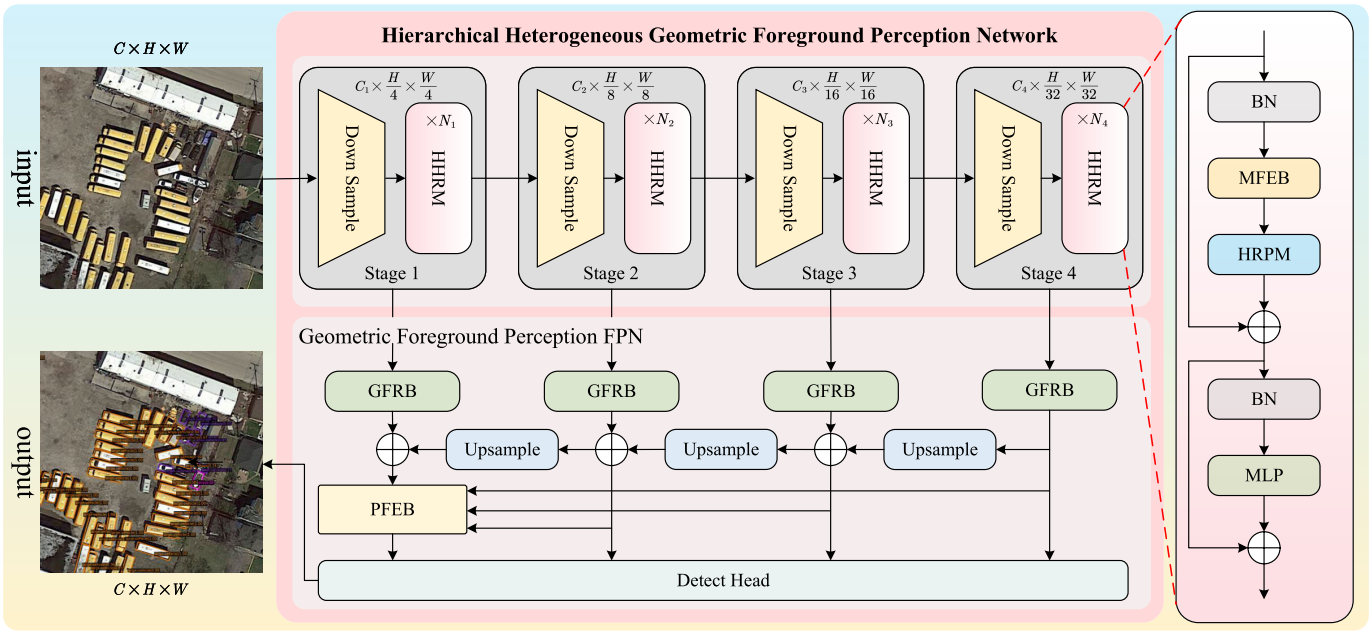

Fig. 1. Overall structure of HHGFP-Net. In HHRM, HRPM utilizes the hierarchical differences of features to reward or penalize the multiscale receptive fields obtained in MFEB, making the features at different levels of the backbone more discriminative. The GFP-FPN can learn the geometric pose of objects via GFRB and further search for and enhance consistency information in multilevel features within PFEB, making the foreground features more prominent.

B. Hierarchical Heterogeneous Receptive-Filed Module

Recent research [9], [11], [13], [21] indicates that integrating large-kernel convolutions and multiscale feature extraction in backbones can enhance the learning of object scale variations in RSIs. This approach has been effectively applied in advanced RSOD frameworks. However, this design can lead to interference between local and global features. Specifically, the broad receptive fields of large convolution kernels can bring global features into shallow layers, marginalizing local features and hindering the detection of small targets. Conversely, local features from smaller kernels can interfere with the global features necessary for detecting large-scale objects in deeper layers, failing to achieve more discriminative representations. To tackle these challenges, this article introduces a HHRM that leverages hierarchical information to enhance critical features and suppress interference at different levels of the backbone, alleviating the limitations of current multiscale feature extraction methods.

The HHRM consists of two batch normalization layers, a multiscale feature extraction block (MFEB), a HRPM, and an MLP layer. The MFEB is used for multiscale feature extraction, while the HRPM complements and suppresses for local or global features during feature fusion by referencing hierarchical information, thus making features at different levels of the backbone more discriminative.

1) Multiscale Feature Extraction Block: MFEB is developed to learn objects of various scales in RSIs using multiscale receptive fields. Given an input feature map $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, we first apply two depthwise convolutions with kernel sizes of 3 and 5, producing two feature maps, denoted as $\{F_S, F_M\} \in \mathbb{R}^{C\times H\times W}$. Then, sum F_S and F_M, and apply a dilated convolution with a kernel size of 7 and a dilation rate of 3 to extract features, resulting in the feature map $F_L \in \mathbb{R}^{C\times H\times W}$. The processing flow of the MFEB can be described as follows:

$$\begin{aligned} F_S &= \text{dwconv}_{3\times 3}(F_{\text{in}}) \\ F_M &= \text{dwconv}_{5\times 5}(F_{\text{in}}) \\ F_L &= \text{dwconv}_{7\times 7,d=3}(F_S + F_M) \end{aligned} \tag{1}$$

where F_S, F_M, and F_L refer to the feature maps obtained under small, medium, and large-scale receptive fields, respectively.

2) Hierarchical Reward and Penalty Mechanism: In this study, the HRPM leverages hierarchical information from the backbone to enhance important features across different layers and penalize interference. The following sections will detail the specifics of HRPM, including the generation of weights for multiscale features in MFEB, the process of mathematical intervention using hierarchical information on these weights, and the final feature fusion method.

In HRPM, spatial weights are used to indicate the importance of various feature scales. To improve computational efficiency, the channels of multiscale features from MFEB are halved

$$\tilde{F}_S, \tilde{F}_M, \tilde{F}_L = \text{conv}_{1\times 1}^{c\to \frac{c}{2}}(F_S, F_M, F_L) \tag{2}$$

where $\{\tilde{F}_S, \tilde{F}_M, \tilde{F}_L\} \in \mathbb{R}^{(c/2)\times H\times W}$ stands for feature map after channel compression.

Subsequently, $\tilde{F}_S$, $\tilde{F}_M$, and $\tilde{F}_L$ are concatenated along the channel dimension. Global average pooling is then applied, followed by MLP layers and a sigmoid activation function to enhance the general representation. Finally, the feature channels are mapped to three and activated by the softmax function. This process can be formally expressed as follows:

$$\begin{aligned} U &= \left[\tilde{F}_S, \tilde{F}_M, \tilde{F}_L\right] \\ U_{\text{SC}} &= \text{mlp}(\text{avgPool}(U)) \end{aligned}$$

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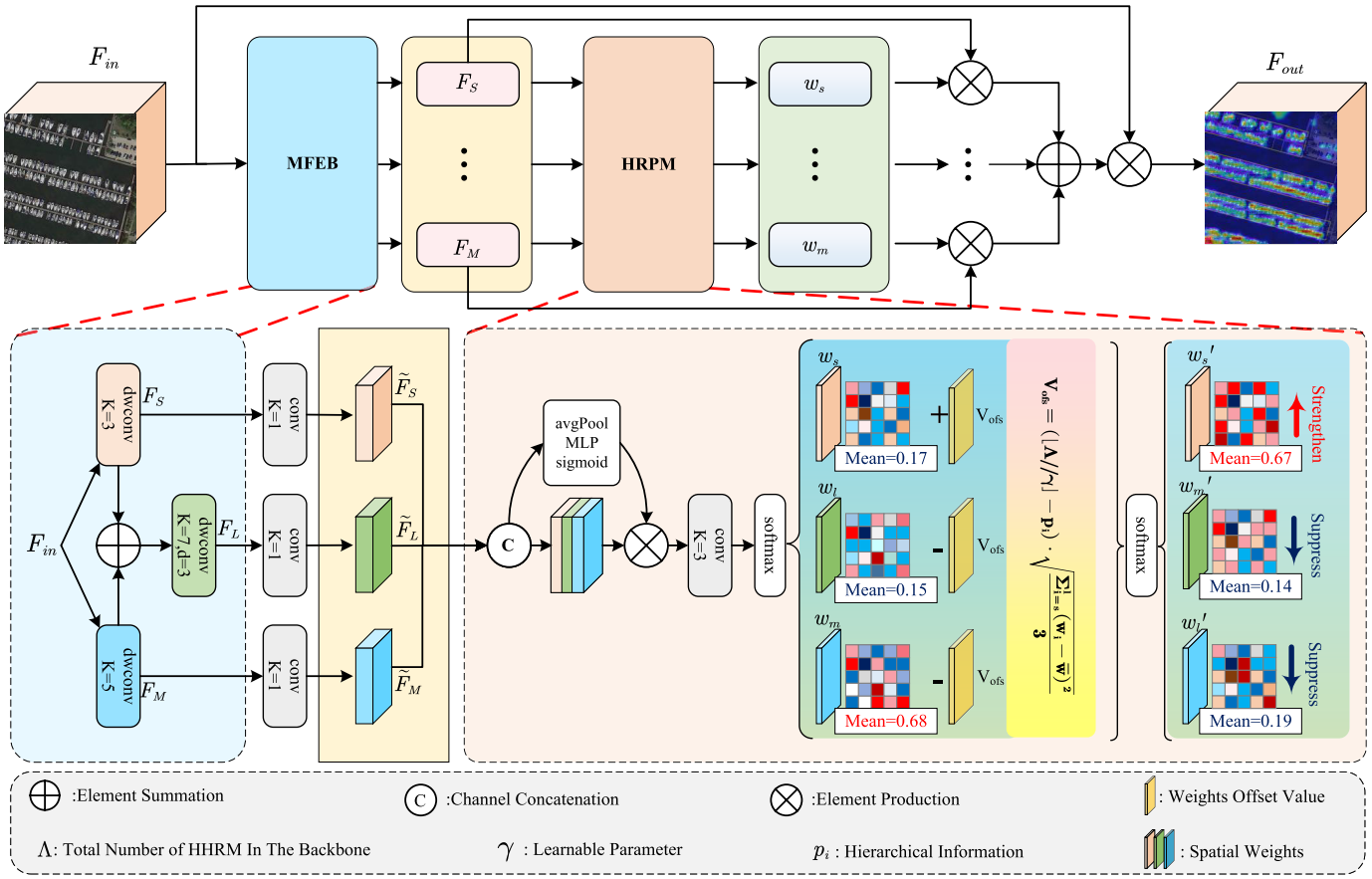

Fig. 2. Overall structure of HHRM. HHRM consists two components: the MFEB and the HRPM. MFEB extracts multiscale features from the input and generates corresponding feature maps. HRPM then evaluates these features to produce spatial weights, integrating hierarchical information to enhance local features in the shallow layers of the backbone while attenuating global features. As the network deepens, global features are gradually reinforced while local features are suppressed, resulting in more discriminative features across different layers of the backbone network.

$$W = U \cdot (\text{sigmoid}(U_{\text{SC}}))$$
$$w_s, w_m, w_l = \text{softmax}\left(\text{conv}_{3\times3}^{C\to3}(W)\right) \tag{3}$$

where w_s, w_m, and w_l stand for the spatial weights of the feature maps $\tilde{F}_S$, $\tilde{F}_M$, and $\tilde{F}_L$, respectively ($w_i \in \mathbb{R}^{1\times H\times W}$, $i \in \{s, m, l\}$), note that the learning of these weights is uncertain due to the black-box nature of deep learning, which may result in the w_l in the shallow layer of the backbone being significantly stronger than the features of other scales, causing interference with local detail information (as shown in Fig. 8).

To leverage hierarchical information in the backbone for supplementing and suppressing features of different scales at corresponding levels, each HHRM in the backbone is assigned an independent positional encoding, denoted as

$$p_i \in [0, \Lambda) \tag{4}$$

where p_i refers to the positional sequence information of the HHRM within the backbone, with $p_i = 0$ for the first occurrence, the symbol Λ denotes the total count of such modules in the backbone.

UniRepLKNet [55] argues the shallow layers of the backbone should prioritize local features and achieve optimal performance by integrating these with global features from deeper layers. However, existing mainstream RSOD methods [11], [13] possess receptive fields of multiple scales, which results in the presence of both local and global features in most backbone layers, hindering the ability to learn discriminative features. To address this issue, we propose HRPM, which uses the hierarchical differences of features to complement for important features and penalize interference information.

To enhance important features across different backbone layers and suppress interference, this article calculates a weight offset and applies it directly to the corresponding feature weights. The offset, denoted as $V_{\text{ofs}} \in \mathbb{R}^{1\times H\times W}$, which is positive in shallow layers of the backbone and negative in deep layers. Its calculation process is as follows:

$$V_{\text{ofs}} = (\lfloor \Lambda//\gamma \rfloor - p_i) \cdot \sqrt{\frac{\Sigma_{i=s}^{l}(w_i - \bar{w})^2}{3}} \tag{5}$$

where γ is a learnable parameter (initialized to 2), and $\bar{w}$ is the average value of w_i, w_m, and w_l over the channel. From the above formula, it can be observed that when the hierarchical information p_i is incorporated into the calculation from the shallow to the deep layers of the backbone, the value of V_{ofs} remains positive in the shallow layers. As the network deepens, its value automatically becomes negative in the deeper layers. By adding V_{ofs} to the weight of local features w_s, it can

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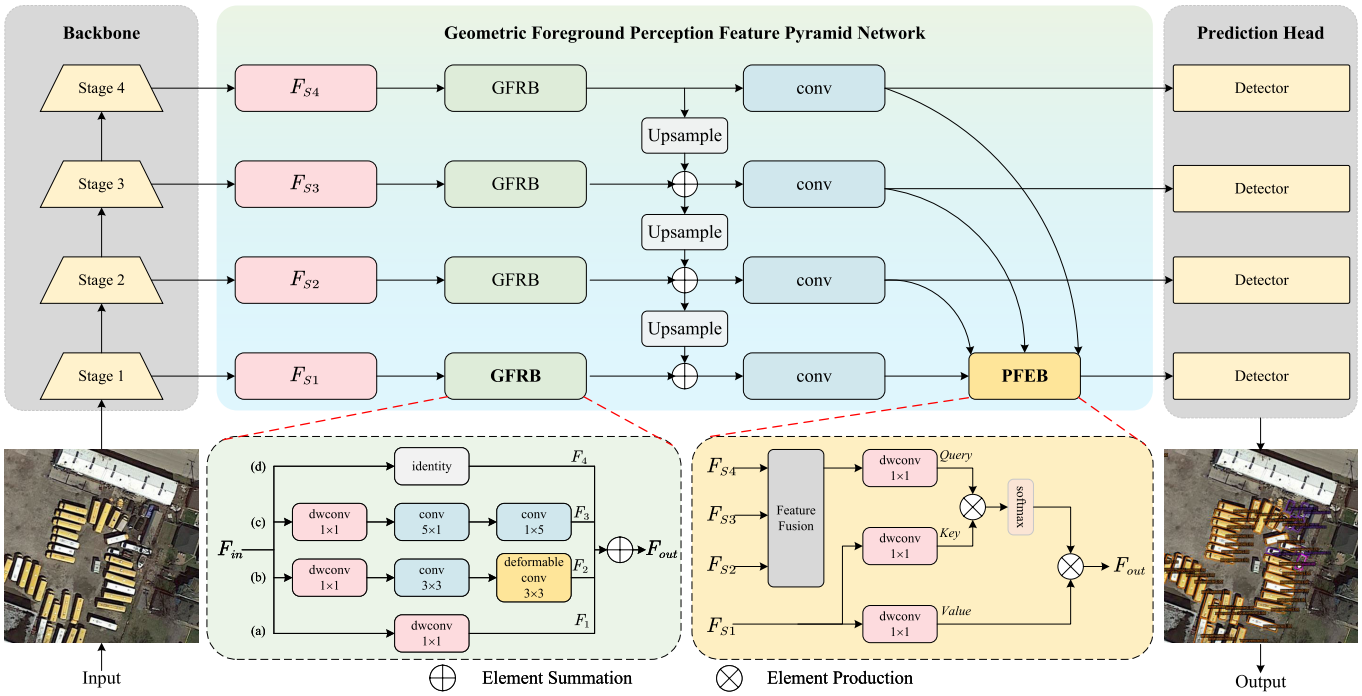

Fig. 3. Overall structure of GFP-FPN. GFP-FPN takes features from different stages of the backbone as input. It first applies the GFRB to each stage's features to refine the geometric posture of objects. Next, the refined features are integrated into the PFEB, which identifies and enhances consistency information. This process improves the prominence of foreground features, leading to more accurate features for final detection.

effectively compensate for local features in the shallow layers and weaken them in the deep layers. Conversely, to suppress global features in the shallower layers and strengthen their representation in the deeper layers, one can subtract V_{ofs} from the initial weights w_m and w_l, reversing the functionality of the formula. It is noteworthy that the offset V_{ofs} is closely related to the position of HHRM within the backbone and is also influenced by the overall weight dispersion. When there is significant disparity in weight distribution across channels, the offset becomes larger. Conversely, if the differences between weights corresponding to different scales are minor, the effect of V_{ofs} is relatively weaker.

Based on the description above, complementing or suppressing features at different scales within the backbone can be achieved using the following formula:

$$\begin{cases} w_s' = \text{softmax}(w_s + V_{\text{ofs}}) \\ w_m' = \text{softmax}(w_m - V_{\text{ofs}}) \\ w_l' = \text{softmax}(w_l - V_{\text{ofs}}) \end{cases} \tag{6}$$

where w_s', w_m', and w_l' refer to the feature weights that have been adjusted based on hierarchical information ($w_i' \in \mathbb{R}^{1\times H\times W}$, $i \in \{s, m, l\}$). These weights correspond, respectively, to the local features $\tilde{F}_S$ with small kernels, and the global features $\tilde{F}_M$ and $\tilde{F}_L$ with large kernels.

The feature fusion and output of the HHRM can be simply expressed as follows:

$$F_{\text{out}} = F_{\text{in}} \cdot \left(w_s' \cdot \tilde{F}_S + w_m' \cdot \tilde{F}_M + w_l' \cdot \tilde{F}_L\right). \tag{7}$$

As illustrated in Fig. 8, the HHRM effectively complements for important features in both shallow and deep layers of the backbone and suppresses interference information during multiscale feature fusion using mathematical interventions. Compared to existing methods, this approach achieves superior local–global feature synergy, which is conducive to learning more discriminative representations.

C. Geometric Foreground Perception-FPN

The neck structure in object detection is a crucial component for interacting with features of different scales. Recently, many neck structures based on variants of FPN [22] have been proposed. Although these models [23], [24], [25], [26] perform well on a variety of downstream tasks, they are primarily optimized for natural images and struggle with RSIs. This issue mainly arises from the inability of previous methods to effectively perceive the posture of remote sensing objects, resulting in less compact features that are more susceptible to background interference. This, in turn, makes the foreground information less distinct and fails to provide more accurate features for final detection, especially for detecting small objects. The GFP-FPN architecture combines geometric modeling with prominent feature enhancement within a unified end-to-end framework. Compared to AO2-DETR [54] and DCFL [15], our approach avoids complex pre- and postprocessing steps while maintaining high computational efficiency.

1) Geometric Feature Refinement Block: Given the randomness and diversity in object positions and orientations within remote sensing imagery, precise feature discrimination is crucial for accurate localization and classification. This article advances the field by introducing a versatile GFRB to enhance the geometric characteristics of objects. The GFRB includes four parallel feature processing branches: Branches (a) and (d)

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preserve the original image features to maintain essential visual information. Branch (b) combines 1×1 depthwise convolution with standard and deformable convolutions to adapt to geometric variations, enhancing the model's ability to handle arbitrary shapes. Branch (c) uses 5×1 and 1×5 strip convolutions [56] to expand the receptive field and reduce computational load, making it effective for detecting linear objects like bridges and ships. This comprehensive feature processing strategy significantly boosts detection accuracy and efficiency in complex RSIs. The formulaic expression of the GFRB processing flow is as follows:

$$\begin{cases} F_1 = \text{dwconv}_{1\times1}(F_{\text{in}}) \\ F_2 = \text{dcn}_{3\times3}(\text{conv}_{3\times3}(\text{dwconv}_{1\times1}(F_{\text{in}}))) \\ F_3 = \text{conv}_{1\times5}(\text{conv}_{5\times1}(\text{dwconv}_{1\times1}(F_{\text{in}}))) \\ F_4 = F_{\text{in}} \end{cases} \tag{8}$$

$$F_{\text{out}} = F_1 + F_2 + F_3 + F_4. \tag{9}$$

2) Prominent Foreground Enhancement Block: Owing to the inherent imaging characteristics of RSIs, object features often are easily obscured by the background, resulting in less prominent foreground features and reduced accuracy. To address this, we propose a PFEB, which identifies and reinforces consistent information across multilevel features, enhancing the prominence of foreground features.

First, the PFEB integrates multilevel features output from the GFRB (denote as F_{Si}) to strengthen prominent consistent features. The fusion process can be expressed by the following formula:

$$\begin{aligned} F_m &= \text{dwconv}_{1\times1}(\text{DS}_{\times4}(F_{S2}) + \text{DS}_{\times2}(F_{S3}) + F_{S4}) \\ F_m &= \text{US}_{\times8}(F_m) \end{aligned} \tag{10}$$

where $\text{DS}_{\times i}$ denotes downsampling by a factor of i, F_m represents the fused features, and $\text{US}_{\times8}$ indicates an eightfold upsampling of the feature map.

In this article, foreground feature enhancement primarily relies on correlation searches among multilevel features. In this process, features with high overlap gain more significant expression due to their consistency, while unstable features, such as background information, are relatively weakened due to their variability. This distinction based on feature stability can be formalized by the following formula:

$$\begin{cases} \text{Query} = \text{dwconv}_{1\times1}(F_m) \\ \text{Value} = \text{dwconv}_{1\times1}(F_{S1}) \\ \text{Key} = \text{dwconv}_{1\times1}(F_{S1}) \end{cases} \tag{11}$$

$$F_{\text{out}} = \text{softmax}(\text{Query} \cdot \text{Key}) \cdot \text{Value} \tag{12}$$

where F_m serving as the Query, engages in a correlation search with F_{S1}, computing the dot product to form an affinity matrix, which is then activated using the Softmax function. This process highlights the foreground features that appear frequently in the feature map, while suppressing the background regions. Multiplying by the Key reduces interference and enhances prominent foreground information. To reduce computational complexity, feature maps for self-attention calculations must undergo shape compression, and should be restored to the original input size after computation. For specific implementation details, please refer to the code we have published.

D. Loss Function

In this article, the loss functions are primarily derived from classification and bounding box regression. The computational representation of the loss function is as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}}. \tag{13}$$

In HHGFP-Net, we used CrossEntropyLoss for classification, denoted as $\mathcal{L}_{\text{cls}}$, and SmoothL1Loss for bounding box regression, indicated by $\mathcal{L}_{\text{reg}}$. The computation of these loss functions is expressed in the following formula:

$$\mathcal{L}_{\text{cls}} = -\sum_{i=1}^{N} y_i \log\left(\hat{y}_i\right) \tag{14}$$

$$\mathcal{L}_{\text{reg}} = \sum \text{smooth}_{L_1}\left(\hat{b}_p - b_p\right), \quad p \in \{x_c, y_c, w, h, \theta\} \tag{15}$$

where N denotes the total number of categories, y_i denotes the one-hot encoding of the true label, and $\hat{y}_i$ predicted probability that the object belongs to category i. For a sample image labeled $y = \{y_i\}_{i=1}^{N}$, $y_i = (b_i, c_i)$ denotes the bounding box b_i and its category label c_i. The prediction set of the algorithm can be denoted as $\hat{y} = \{\hat{y}\}_{i=1}^{N}$ where $\hat{y}_i = (\hat{b}_i, \hat{c}_i)$. Since it is a prediction of a rotated box, it is also necessary to introduce a rotational angle $\theta \in (0, \frac{\pi}{2}]$.

IV. Experiments and Analysis

We conducted extensive experiments on four publicly available RSOD datasets to validate the effectiveness of the proposed HHGFP-Net. The datasets and evaluation metrics we used will be introduced in Sections IV-A and IV-B, respectively. Detailed experimental specifics are provided in Section IV-C. Comparison results with the best-performing methods on each dataset are provided in Section IV-D. Finally, in Section IV-E, we deliver an ablation study on the proposed methods.

A. Datasets

DOTA-v1.0 [12] is a large RSOD dataset containing over 2800 aerial images, encompassing more than 180 000 labels across 15 fine-grained categories, including airplanes (PL), ships (SH), and small vehicles (SVs). The DOTA dataset only provides researchers with true annotations for the training and testing sets, while the test set provides only images that require inference results to be uploaded to a server for online validation.

DOTA-v1.5 [12] is based on the DOTA-v1.0 release, with images consistent between the two versions. However, DOTA-v1.5 adds more annotations for small objects less than 10 pixels and introduces a new category for container crane (CC). The total number of instance labels exceeds 400 000, enabling experiments to better validate performance in detecting small objects.

DIOR-R [28] is a RSOD dataset that adds OBB annotations to the DIOR dataset. It contains over 23 000 satellite images

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and nearly 200 000 instances, covering 20 common fine-grained categories, including airplanes, baseball fields, and ships. Detecting objects in DIOR-R is particularly challenging due to the presence of more small objects and more distracting backgrounds.

STAR [29] is a large-scale dataset for scene graph generation. It is primarily used to understand spatial relationships between objects in RSIs. Since object detection is fundamental task to scene graph generation, the STAR dataset offers over 210 000 instances and 4000 triplet labels describing spatial relationships, covering 48 fine-grained categories such as ships, vehicles, airplanes, and bridges.

B. Evaluation Metric

In this article, MaxIoU is adopted as the label assignment strategy, for all the anchor boxes obtained from the prediction, we first calculate the maximum IoU corresponding to the GT that matches it the most, and the calculation of IoU can be described by the following formula:

$$\text{IoU} = \frac{\text{Area}(B \cap G)}{\text{Area}(B \cup G)} >= \text{Threshold} \tag{16}$$

where B denotes the detected bounding box and G denotes the ground truth. A predicted object is considered a true positive (TP) when its intersection over union (IoU) is larger than the set threshold, and a false positive (FP) when it is less than the threshold. Ground truth objects that fail to be matched are referred to as false negatives (FNs). From this, the precision (P) and recall (R) can be further calculated, and the calculation process can be formulated as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{18}$$

We adopted average precision (AP) and mean AP (mAP) as the main metrics for evaluating performance on RSOD. AP50 in the subsequent experimental section indicates that the threshold is set to 0.5, and N denotes the number of species to be predicted, which can be expressed as

$$\text{AP} = \int_0^1 P(R)d(R) \tag{19}$$

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N}\int_0^1 P_i(R_i)dR_i. \tag{20}$$

C. Implementation Details

In this study, our training strategy aligns with the mainstream approaches, utilizing the training and validation sets for model training and the test set for model performance evaluation. We preprocessed the original images of DOTA-v1.0, DOTA-v1.5, and STAR by cropping them with scaling rates of 0.5, 1.0, 1.5, fixing overlapping pixels at 500, while no additional processing was performed on DIOR-R. Our visual backbone was initially pretrained on ImageNet for 300 epochs, followed by fine-tuning on the downstream dataset for 12 epochs. We utilized AdamW as the optimizer, with a weight decay of 0.05 and a learning rate of 0.0001. The input image size of 1024×1024 was used for all FLOPs calculations to ensure accurate evaluations. Our experiments were conducted on 5 RTX 3090 GPUs, and the code is implemented based on the MMRotate and PyTorch.

D. Comparisons With Previous Methods

We conducted extensive comparative experiments on four public datasets to validate the effectiveness of HHGFP-Net. Furthermore, we provide visual qualitative comparison results with suboptimal methods for each dataset to illustrate the performance of our approach in various challenging scenarios.

1) Quantitative and Qualitative Result on DOTA-v1.0 Dataset: Table I reports the performance comparison of our proposed HHGFP-Net with mainstream methods on the DOTA-v1.0 dataset. With the inclusion of EMA tuning techniques, our method significantly outperforms other oriented detectors with an mAP of 80.68%. Compared to the state-of-the-art FPNFormer [47] and PKINet [11], we achieve improvements of 0.93% and 0.7%, respectively. Especially in fine-grained detection, HHGFP-Net achieved the best AP in several categories such as "baseball diamond (BD)," "bridge (BR)," and "ground-track field (GTF)."

To investigate the underlying causes of these results, we analyzed the contextual modeling capabilities and algorithmic distinctions between HHGFP-Net and other state-of-the-art multiscale methods. Specifically, PKINet employs element-wise addition to fuse multiscale features, which can result in the coexistence of features from different scales at the same level, causing mutual interference and diminishing the clarity of discriminative features. As demonstrated in the qualitative comparison in Fig. 4, PKINet easily misdetects small objects or exhibits mismatches in complex backgrounds. The large receptive fields further introduces background noise into the recognition process, thereby hindering the detection of small objects. In contrast, HHGFP-Net enhances local features in shallow layers to improve the detection of small objects, while simultaneously strengthening global features in deep layers to facilitate the detection of large objects (Fig. 9). This dual enhancement mechanism enables HHGFP-Net to achieve accurate small object detection at shallow levels and maintain strong performance for large objects at deeper levels.

2) Quantitative Result on DOTA-v1.5: The DOTA-v1.5 dataset incorporates additional annotations for small objects, making it well-suited for evaluating a model's performance in small object detection. Table II and Fig. 5 present the comparison results of HHGFP-Net against seven top-performing methods. HHGFP-Net achieves the highest mAP of 77.54% on this dataset. In detecting small objects such as "SV," "large vehicles (LV)," and "ships (SH)," methods utilizing multiscale feature extraction, including LSKNet, PKINet, and HHGFP-Net, perform worse than REDet [40]. REDet uses ResNet [17] as the backbone network, which excels at capturing local features, a capability crucial for small object detection. In comparison, LSKNet and PKINet are susceptible to local information interference by global features during shallow-layer small object detection,

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TABLE I

EXPERIMENTAL RESULTS ON THE **DOTA-v1.0** DATASET. RESULTS LABELED IN **RED**, **BLUE**, AND **GREEN** INDICATE THE BEST, SECOND BEST, AND THIRD BEST PERFORMANCE IN EACH COLUMN

Method	Backbone	Venue	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
SCRDet [14]	R101	ICCV2019	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	86.86	65.02	66.68	66.25	68.24	65.21	72.61
FAOD [41]	R101	ICIP2019	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.84	73.28
Gliding Vertex [16]	R101	TPAMI2020	89.64	**85.00**	52.26	**77.34**	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
CenterMap [57]	R50	TGRS2020	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.70	71.74
CSL [58]	R152	ECCV2020	**90.25**	**85.53**	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93	76.61
FR-Est [59]	R101	TGRS2020	89.63	81.17	50.44	70.19	73.52	77.98	86.44	90.82	84.13	83.56	60.64	66.59	70.59	66.72	60.55	74.20
S^2ANet [60]	R50	TGRS2021	89.30	80.11	50.97	73.91	78.59	77.34	86.38	**90.91**	85.14	84.84	60.45	66.94	66.78	68.55	51.65	74.13
REDet [40]	R101	CVPR2021	89.80	82.90	48.60	65.20	69.50	70.10	70.20	90.50	85.60	83.40	62.50	63.90	65.60	67.20	68.00	72.20
O-RCNN [36]	R101	ICCV2021	89.51	84.50	58.67	73.10	79.67	84.87	88.08	**90.90**	**86.97**	85.38	68.06	67.35	75.91	78.83	58.91	78.04
SASM [61]	R50	AAAI2022	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92
G-Rep [37]	R101	RS2023	88.89	74.62	43.92	70.24	67.26	67.26	79.80	90.87	84.46	78.47	54.59	62.60	66.67	67.98	52.16	70.59
ARC [27]	ARC-R101	ICCV2023	89.39	83.58	57.51	75.94	78.75	83.58	88.08	90.86	85.93	85.38	64.03	68.65	75.59	72.03	65.68	77.70
FPNFormer [47]	R50	TGRS2023	89.90	84.61	**58.74**	**81.96**	79.94	84.22	**89.11**	90.77	84.88	**87.91**	**70.31**	68.23	**78.33**	77.71	69.56	**79.75**
ASDet [62]	R50	TGRS2024	**90.34**	83.94	55.51	71.10	**82.00**	84.71	**89.29**	**90.90**	85.71	**87.26**	62.84	68.65	76.65	**80.34**	65.21	78.30
HOFA-Net [63]	CSPDarknet53	JSTARS2024	**90.42**	76.64	47.57	59.01	73.41	**85.64**	**89.29**	90.76	73.30	**89.44**	**71.15**	69.39	75.16	67.05	**74.77**	75.53
ABFL [64]	R-152	TGRS2024	89.73	83.85	56.56	70.78	**81.85**	84.61	88.71	90.85	85.74	86.50	67.91	**71.28**	**78.61**	**80.69**	65.56	78.88
PKINet [11]	PKINet-S	CVPR2024	88.57	81.50	**61.06**	76.75	**80.12**	**85.25**	89.10	90.63	**87.14**	86.91	69.80	**72.02**	77.83	**83.47**	**69.60**	**79.98**
HHGFP-Net(Ours)	VAN [21]+HHRM	-	89.60	**85.72**	**61.36**	**82.11**	78.91	**85.90**	88.55	90.09	86.81	86.48	**70.93**	68.59	**78.64**	78.81	**76.82**	**80.68**

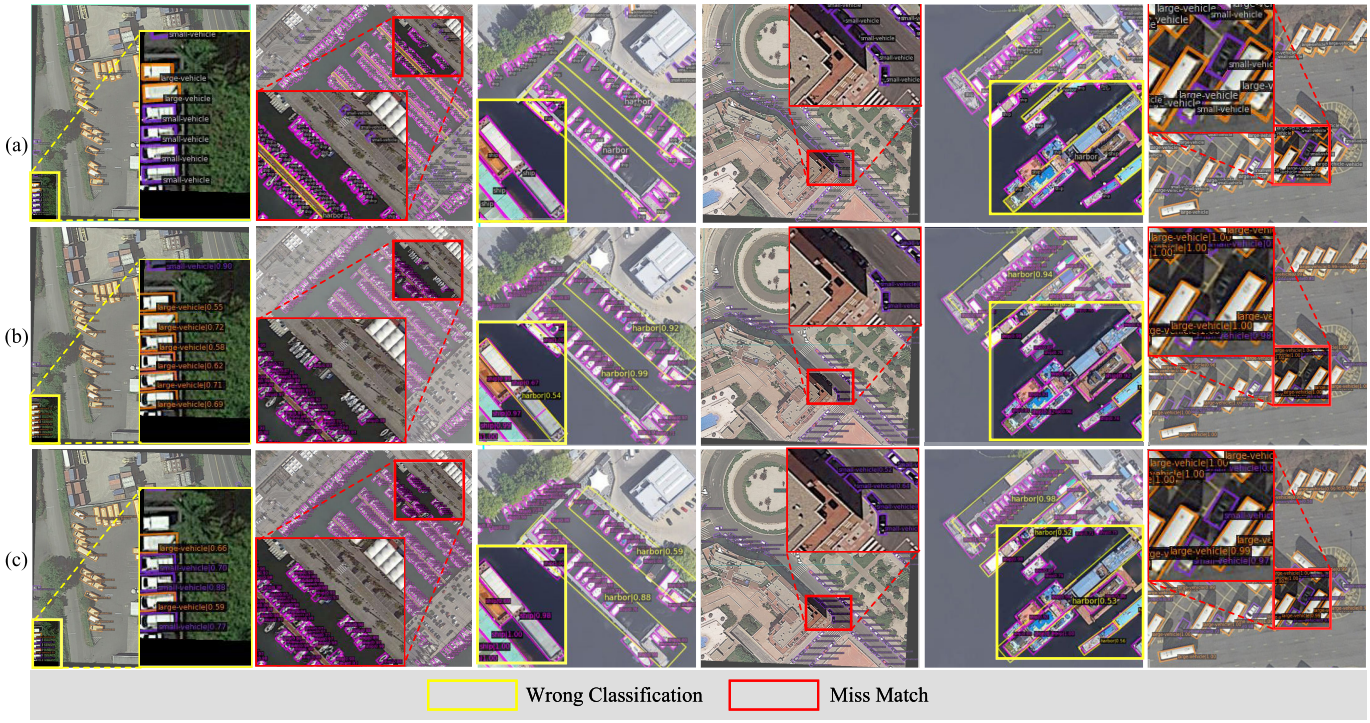

Fig. 4. Qualitative comparisons on DOTA-v1.0 validation. The top row shows the ground-truth annotations for reference. The second row displays results from PKINet, while the last row presents results from our proposed HHGFP-Net. Different categories are differentiated by the color of the bounding boxes. We used a threshold of 0.5 to filter out bounding boxes with low confidence. Additionally, we reduced the transparency of regions with insignificant differences in the images to better illustrate the superiority of our method. (a) GT. (b) PKINet. (c) HHGFP-Net.

leading to suboptimal performance. HHGFP-Net mitigates this issue by strengthening the representation of local feature branches in shallow layers while suppressing global feature interference, thereby improving small object detection performance. Compared to LSKNet and PKINet, HHGFP-Net achieves improvements of 5.03% and 5.04%, respectively, in detecting "SV." Furthermore, for larger objects such as "Roundabouts," HHGFP-Net delivers superior performance, attributed to its enhanced contextual awareness in deep layers and its effective suppression of local feature interference.

3) Quantitative Result on DIOR-R: Table III and Fig. 6 show that HHGFP-Net achieves excellent performance on the DIOR-R dataset, with an mAP of 72.34%. Among the 20 subcategories, HHGFP-Net ranks in the top three for 19 classes. For objects with extreme aspect ratios, such as "bridges (BR)," "ships (SH)," and "harbors (HA)," The reason for this result is fully explored in ablation experiments, which demonstrate that this advantage stems from the irregular multiscale feature encoding and enhancement mechanism of GFP-FPN, which can efficiently capture the spatial and

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TABLE II

EXPERIMENTAL RESULTS ON **DOTA-v1.5** DATASET. RESULTS LABELED IN **RED**, **BLUE**, AND **GREEN** INDICATE THE BEST, SECOND BEST, AND THIRD BEST PERFORMANCE IN EACH COLUMN

Method	Backbone	**Venue**	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	**mAP**
Rotated RetinaNet [65]	R152	ICCV2017	71.43	77.64	42.12	64.65	44.53	56.79	73.31	**90.84**	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
FR OBB [12]	R50	CVPR2018	71.89	74.47	44.45	59.87	51.28	69.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
HTC [48]	R50	CVPR2019	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
REDet [40]	R101	CVPR2021	**88.51**	**86.45**	**61.23**	81.20	**67.60**	**83.65**	**90.00**	**90.84**	**84.30**	75.33	**71.49**	72.06	**78.32**	**74.73**	**76.10**	46.98	**76.80**
RHINO [66]	Swin-T	CVPR2023	79.96	**85.16**	56.96	74.50	**64.14**	**81.91**	**89.42**	90.60	**83.93**	**82.89**	59.93	**74.84**	77.23	**75.18**	64.91	33.76	73.46
LSKNet [13]	LSKNet-S	ECCV2023	72.15	84.91	**60.44**	**84.93**	62.42	77.45	81.07	**90.85**	79.44	**79.00**	**72.10**	**73.65**	**77.49**	**75.39**	**75.81**	**52.19**	**74.95**
PKINet [11]	PKINet-S	CVPR2024	**80.52**	**85.23**	55.25	**84.28**	62.41	76.81	88.37	90.77	79.24	68.48	67.87	72.45	76.24	74.62	73.97	**57.23**	74.56
HHGFP-Net(Ours)	VAN [21]+HHRM	-	**80.75**	84.10	**60.81**	**83.22**	**67.45**	**83.55**	**89.78**	**90.88**	**84.86**	**77.63**	**71.75**	**76.17**	**77.34**	73.15	**81.11**	**58.13**	**77.54**

Fig. 5. Qualitative comparisons on DOTA-v1.5 validation. The top row shows the ground-truth annotations for reference. The second row displays results from REDet, while the last row presents results from our proposed HHGFP-Net. Important areas in the image have been highlighted with red and yellow borders, while the transparency of surrounding areas has been reduced. (a) GT. (b) REDet. (c) HHGFP-Net.

TABLE III

EXPERIMENTAL RESULTS ON **DIOR-R** DATASET. RESULTS LABELED IN **RED**, **BLUE**, AND **GREEN** INDICATE THE BEST, SECOND BEST, AND THIRD BEST PERFORMANCE IN EACH COLUMN

Method	**Backbone**	**Venue**	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	**mAP**
Rotated RetinaNet [65]	R152	ICCV2017	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	67.57	61.57	81.46	47.33	38.01	60.24	57.55
FR OBB [12]	R50	CVPR2018	62.79	26.80	71.72	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	**71.21**	81.44	47.31	50.46	65.21	59.54
ROI Trans [67]	R50	CVPR2019	63.12	25.34	71.83	81.33	36.04	72.62	20.33	75.32	68.44	68.92	82.74	37.72	52.73	81.33	73.42	70.71	81.53	45.32	**51.12**	**66.24**	61.30
CSL [58]	R50	ECCV2020	68.02	25.33	76.12	81.40	21.33	72.71	19.13	73.30	54.74	74.32	77.92	29.42	37.62	79.44	68.52	64.32	81.44	44.83	41.34	63.62	57.74
Gliding Vertex [16]	R50	TPAMI2020	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
R3Det [68]	R101	AAAI2021	67.10	41.63	74.34	81.33	35.61	74.72	20.82	76.33	62.34	72.31	78.14	31.92	52.81	80.42	70.43	68.63	81.54	51.51	45.52	63.93	61.55
O-RCNN [36]	R50	ICCV2021	63.13	36.73	72.11	81.32	44.33	72.72	26.94	**77.70**	67.91	68.72	83.11	40.93	57.74	**89.51**	73.65	**71.13**	81.41	53.81	51.10	66.10	64.05
RVSA [69]	ViTAE-B+RVSA	TGRS2022	**81.20**	**54.71**	**81.12**	**88.13**	**51.83**	**79.93**	**36.79**	76.06	**89.23**	**78.30**	**84.46**	47.29	**65.01**	81.19	**82.17**	70.69	**90.03**	**66.75**	50.73	65.40	**71.05**
DODet [62]	R50	TGRS2023	63.40	43.35	72.11	81.32	43.12	72.59	33.32	**78.77**	70.84	74.15	75.47	**48.00**	59.31	**85.41**	**74.04**	**71.56**	81.52	55.47	**51.86**	**66.40**	65.10
MTP [42]	InternImage-XL	JSTARS2024	**89.61**	**52.64**	**81.23**	**87.82**	**48.60**	**77.21**	**43.32**	71.60	**89.13**	**79.04**	**84.02**	**51.32**	**60.91**	81.02	**83.37**	71.12	**90.02**	**66.61**	50.53	66.00	**71.30**
HHGFP-Net(Ours)	VAN [21]+HHRM	-	**72.03**	**52.04**	**80.73**	**89.63**	**53.42**	**81.33**	**39.15**	**88.42**	**79.31**	**78.62**	**84.72**	**52.03**	**64.02**	**90.01**	**80.93**	71.12	**89.93**	**65.82**	**58.73**	**74.46**	**72.34**

geometric properties of the target before the final detection phase. On several other datasets used in this article, HHGFP-Net can also present advantages in detecting such objects.

4) Quantitative Result on STAR: The STAR dataset features over 40 fine-grained categories, necessitating stringent discriminative representation. The quantitative results in Table IV demonstrate that HHGFP-Net outperforms other methods significantly, achieving an mAP of 39.9%. Notably, HHGFP-Net exhibits exceptional precision for fine-grained object detection, such as ships, bridges, and gravity dams,

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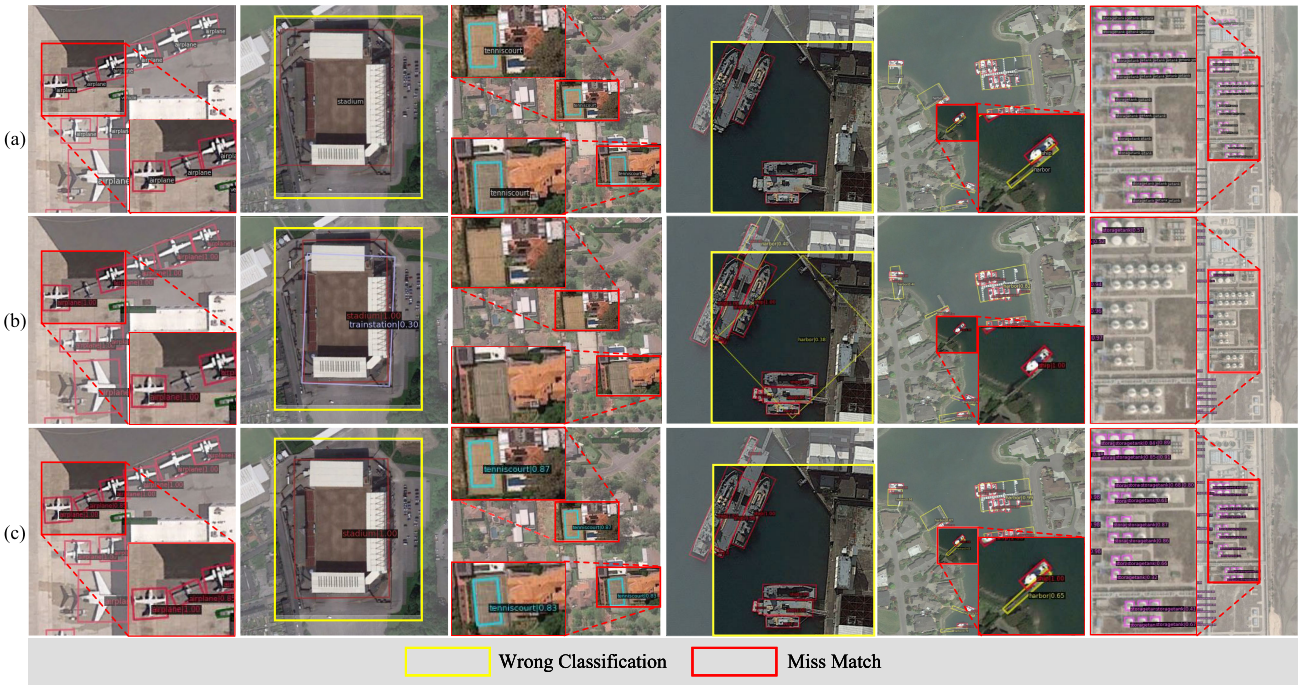

Fig. 6. Qualitative comparisons on DIOR-R validation. The top row shows the ground-truth annotations for reference. The second row displays results from MTP, while the last row presents results from our proposed HHGFP-Net. We distinguish different categories by the color of the bounding boxes. Furthermore, we decreased the visibility of areas with negligible differences in the images to more clearly highlight the benefits of our method. (a) GT. (b) MTP. (c) HHGFP-Net.

TABLE IV

EXPERIMENTAL RESULTS ON **STAR** DATASET. RESULTS LABELED IN **RED**, **BLUE**, AND **GREEN** INDICATE THE BEST, SECOND BEST, AND THIRD BEST PERFORMANCE IN EACH COLUMN

Method	Backbone	Venue	ship	boat	truck	car	airplane	crane	b_b	tank	l_t	bridge	runway	s_l	dock	g_d	…	mAP
RoI Trans [67]	R50	CVPR2019	56.6	**26.2**	50.3	**69.1**	**89.9**	59.7	**63.4**	**57.3**	**51.0**	15.8	0.0	0.0	13.6	0.0	…	**35.7**
G.V [16]	R101	TPAMI2020	50.6	23.8	38.6	68.1	89.4	56.9	54.7	54.2	45.9	17.8	0.0	0.0	14.9	0.0	…	30.7
O-RCNN [36]	R101	ICCV2021	**57.6**	**26.6**	**51.6**	67.8	89.5	**65.0**	**62.8**	54.1	49.1	**21.2**	0.0	0.0	14.8	0.1	…	33.2
GCL [70]	R101	KBS2023	30.6	7.2	26.6	38.1	54.3	23.5	24.5	21.1	21.8	5.4	**6.6**	0.0	5.0	2.2	…	34.4
GWD [49]	R101	ICML2023	42.5	14.5	38.4	65.8	89.1	52.2	54.0	51.0	49.2	4.4	0.0	0.0	12.6	0.0	…	25.3
DCFL [15]	R50	CVPR2023	47.3	10.2	45.6	**69.7**	89.2	56.8	52.9	**55.2**	49.2	13.2	0.0	0.0	11.3	2.3	…	29.0
KFIoU [71]	R50	ICLR2023	41.6	14.1	39.5	67.1	89.4	47.7	54.6	51.4	47.0	8.3	0.0	0.0	9.2	**4.5**	…	25.5
LSKNet [13]	LSKNet-S	ECCV2023	**59.0**	25.5	**53.2**	**70.9**	89.7	**65.5**	**62.4**	53.6	**55.0**	**24.7**	0.0	0.0	**19.5**	**11.4**	…	**37.8**
ARS-DETR [72]	Swin-T	TGRS2024	44.6	16.4	36.1	60.8	88.6	40.5	59.0	54.7	41.6	8.2	0.0	0.0	7.8	0.0	…	28.1
PSC [73]	R101	TPAMI2024	44.6	18.0	44.3	66.2	89.6	56.4	55.4	53.1	44.8	16.1	0.0	0.0	13.9	0.0	…	30.5
PKINet [11]	PKINet-S	CVPR2024	55.2	22.4	48.0	67.5	**89.9**	59.1	56.3	50.7	47.4	21.1	0.0	0.0	**19.0**	0.2	…	32.8
HHGFP-Net(Ours)	VAN [21]+HHRM	-	**59.6**	**32.0**	**53.9**	60.2	**90.1**	**68.1**	61.1	**58.2**	**57.9**	**27.0**	0.0	0.0	**21.5**	**34.6**	…	**39.9**

which typically have high aspect ratios. These object characteristics are effectively learned within the GFRB, leading to improved detection performance. On the STAR dataset, where complex backgrounds and indistinct foregrounds are prevalent, traditional methods often fail to suppress background interference, resulting in detection failures (Fig. 7). To address this, the PFEB module performs cross-layer searches for consistent information across different levels of features, enhancing texture information. As shown in Fig. 11, GFP-FPN combined with PFEB effectively reduces background noise, making the foreground more prominent and enabling more precise detection in these challenging scenes.

E. Ablation Study

We conducted extensive ablation experiments to validate the effectiveness of each component in our proposed HHGFP-Net. As previously stated, our backbone network is constructed based on HHRM and VAN, therefore, we denote our visual backbone as "VAN + HHRM" in this article.

1) Effectiveness of Individual Component: We conducted ablation studies on each key component of HHGFP-Net. As shown in Table VI, both HHRM and GFP-FPN have been proven beneficial for improving detection performance. Specifically, when O-RCNN uses ResNet and FPN, the mAP reaches 78.04%, and using VAN as the backbone can achieve a significant improvement of 0.88% compared to ResNet. Based on these results, we embed HHRM as a core component into VAN, achieving an optimal performance of 79.83%. Additionally, in the ablation experiments regarding GFP-FPN, we found that it can provide varying degrees of accuracy improvement compared to traditional FPN. When combining "VAN + HHRM," we observed the best inference results,

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Fig. 7. Qualitative comparisons on STAR validation. The top row shows the ground-truth annotations for reference. The second row displays results from LSKNet, while the last row presents results from our proposed HHGFP-Net. Different colored bounding boxes represent objects from various fine-grained categories. We also lowered the transparency of less significant regions in the images to more effectively showcase the advantages of our HHGFP-Net. (a) GT. (b) LSKNet. (c) HHGFP-Net.

TABLE V

QUANTITATIVE COMPARISON RESULTS ON THE SINGLE-SCALE CROPPED DOTA-V1.0 DATASET. BOLD TEXT REPRESENTS THE BEST PERFORMANCE IN EACH COLUMN

Method	Venue	#Params	FPS	mAP
SCRDet [14]	ICCV2019	55.1M	12.2	72.61
CenterMap [57]	TGRS2020	41.1M	11.6	71.74
G.V [16]	TPAMI2020	41.1M	13.3	75.02
REDet [40]	CVPR2021	47.4M	13.4	72.20
R3Det [68]	AAAI2021	41.9M	14.7	73.79
S^2ANet [60]	TGRS2021	38.6M	14.6	74.13
O-RCNN [36]	ICCV2021	41.1M	12.7	78.04
FPNFormer [47]	TGRS2023	34.3M	14.1	79.75
PKINet [11]	CVPR2024	**30.8M**	**16.4**	79.98
HHGFP-Net(Ours)†	-	37.6M	13.2	**80.68**

TABLE VI

ABLATION STUDY OF THE FUNDAMENTAL COMPONENTS OF HHGFP-NET ON THE DOTA-V1.0 DATASET. BOLD TEXT REPRESENTS THE BEST PERFORMANCE

	Backbone			Neck		mAP
	ResNet101	VAN	VAN+HHRM	FPN	GFP-FPN	
O-RCNN	✓			✓		78.04
		✓		✓		78.92
			✓	✓		79.83
	✓				✓	78.45
		✓			✓	79.69
			✓		✓	**80.68**

TABLE VII

ABLATION STUDY ON THE SETTING OF MFEB RECEPTIVE FIELDS ON THE DOTA-V1.0 DATASET. BOLD TEXT REPRESENTS THE BEST PERFORMANCE IN EACH COLUMN

Module Implementation	Max RF	FLOPs(G)	mAP
(k=3,d=1),(k=5,d=1)	7	**36.60**	62.32
(k=5,d=1),(k=7,d=3)	23	37.31	63.57
(k=3,d=1) → (k=5,d=1) → (k=7,d=3)	25	38.86	63.31
(k=3,d=1),(k=5,d=1)→ (k=7,d=3)	23	38.86	**64.50**

reaching 80.68%, outperforming state-of-the-art mainstream methods [11], [13], [27], [62].

2) Ablation Experiments on HHRM: In HHRM, we first explored the optimal receptive field setting strategy in MFEB (Table VII). Next, we investigated the role of the HRPM in mitigating interference among multiscale features within the backbone (Fig. 8). Finally, we incorporated HHRM into the VAN as the backbone network and embedded it into several mainstream methods for training and testing, with experimental results demonstrating that HHRM effectively enhances performance in RSOD (Table VI).

The setting of the receptive field is crucial for learning features of objects at different scales. Therefore, our ablation experiments on MFEB focus on determining the optimal

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TABLE VIII

ABLATION STUDY OF THE FUNDAMENTAL COMPONENTS OF GFP-FPN ON THE DOTA-V1.0 DATASET. BOLD TEXT REPRESENTS THE BEST PERFORMANCE IN EACH COLUMN

Neck	GFRB	PFEB	#P(M)	FLOPs(G)	mAP
FPN			**29.42**	**167.48**	79.83
GFP-FPN	✓		33.52	173.89	80.15
		✓	34.01	175.69	80.10
	✓	✓	37.61	180.46	**80.68**

TABLE IX

ABLATION STUDY OF THE EFFECT OF GFP-FPN COMPONENTS ON SMALL OBJECTS AND OBJECTS WITH EXTREME ASPECT RATIOS ON STAR

Method	ship	bridge	boat	g_d	mAP
FPN	55.3	23.9	30.1	14.5	35.8
FPN+GFRB	57.8	26.3	**33.1**	**34.8**	37.6
FPN+GFRB+PFEB	**59.6**	**27.0**	32.0	34.6	**39.9**

TABLE X

DETECTION RESULTS FOR TYPICAL ARBITRARY ORIENTATION AND EXTREME SHAPE OBJECTS ON STAR

Method	ship	bridge	g_d	mAP
AO2DETR [54]	56.4	24.2	24.5	36.2
GCWNet [45]	54.7	25.8	27.5	29.4
DCFL [15]	57.8	26.9	32.0	37.8
HHGFP-Net(Ours)†	**59.6**	**27.0**	**34.6**	**39.9**

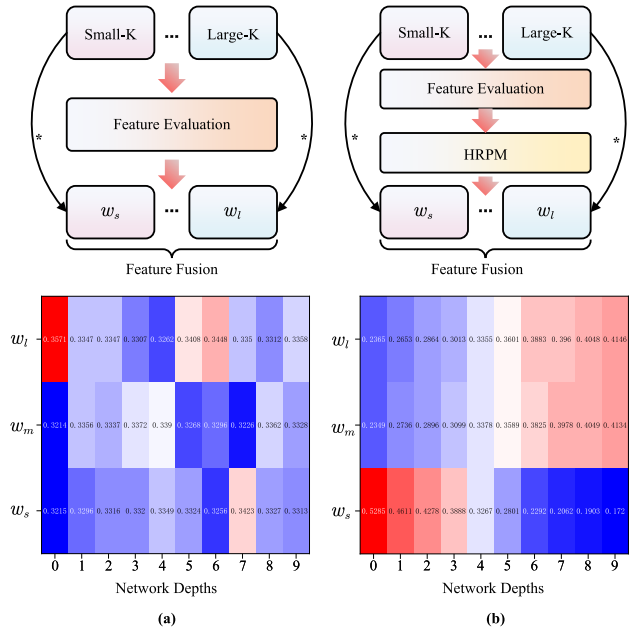

Fig. 8. Results of the ablation study of HHRM demonstrate that applying HRPM can effectively mitigate interference between features at different levels of the backbone, achieving more discriminative representations. Green and red bounding boxes represent TP and FN prediction results, respectively. "K": Kernel. "w_s, w_m, w_l": Feature weights under small, medium, and large receptive filed. (a) Before HRPM. (b) After HRPM.

receptive field size for multiscale feature extraction. The results of these experiments are presented in Table VII. Extensive testing revealed that configuring the feature extraction architecture with three types of receptive fields, with a maximum size of 23, yields the best accuracy.

TABLE XI

ABLATION STUDY OF EMBEDDING THE BASIC COMPONENTS OF HHGFP-NET INTO OTHER METHODS

Method	Backbone	Neck	#P(M)	FLOPs(G)	mAP(%)
R3Det	ResNet50	FPN	41.92	336	76.47
R3Det	ResNet50	GFP-FPN	98.47	442	77.52 (↑1.05)
R3Det	VAN+HHRM	FPN	26.81	297	77.12 (↑0.65)
R3Det	VAN+HHRM	GFP-FPN	33.72	398	77.92 (↑1.45)
CenterMap	ResNet50	FPN	41.14	198	76.03
CenterMap	ResNet50	GFP-FPN	84.34	388	77.23 (↑1.20)
CenterMap	VAN+HHRM	FPN	55.58	241	76.12 (↑0.09)
CenterMap	VAN+HHRM	GFP-FPN	61.52	320	77.49 (↑1.46)
G.V.	ResNet101	FPN	41.15	198	75.02
G.V.	ResNet101	GFP-FPN	96.21	305	75.81 (↑0.79)
G.V.	VAN+HHRM	FPN	28.16	163	75.77 (↑0.75)
G.V.	VAN+HHRM	GFP-FPN	37.61	180	76.12 (↑1.10)
ROI Trans	ResNet101	FPN	55.13	200	74.61
ROI Trans	ResNet101	GFP-FPN	98.56	326	75.23 (↑0.62)
ROI Trans	VAN+HHRM	FPN	43.44	181	74.73 (↑0.12)
ROI Trans	VAN+HHRM	GFP-FPN	51.59	194	75.68 (↑1.07)
CFA	ResNet152	FPN	71.23	350	76.67
CFA	ResNet152	GFP-FPN	125.11	484	77.85 (↑1.18)
CFA	VAN+HHRM	FPN	21.39	153	77.56 (↑0.89)
CFA	VAN+HHRM	GFP-FPN	28.42	426	78.03 (↑1.36)
O-RCNN	ResNet101	FPN	41.1	211	78.04
O-RCNN	ResNet101	GFP-FPN	97.66	336	78.45 (↑0.41)
O-RCNN	VAN+HHRM	FPN	29.42	167	79.83 (↑1.79)
O-RCNN	VAN+HHRM	GFP-FPN	37.61	180	80.68 (↑2.64)

Existing multiscale feature extraction methods can be broadly categorized into two types for feature fusion: one involves directly performing pixel-wise addition of features from different scales, while the other uses attention mechanisms to weight features before soft fusion. However, both approaches can lead to interference among features at different levels of the backbone [see Fig. 8(a)]. To address this issue, we propose HRPM, which complements for important features and suppresses interference across different backbone levels. The ablation results for HRPM are presented in Fig. 8(b). The results indicate that our proposed method enhances local features in shallow layers with small receptive fields, while emphasizing global features with larger receptive fields in deeper layers. This strategy effectively reduces interference between features, making multiscale feature representations at different levels more discriminative.

To further validate the effectiveness of HHRM, we integrated "VAN [21] + HHRM" as the backbone network and evaluated it across several mainstream algorithms. Experimental results presented in Table XI demonstrated that HHRM significantly enhances accuracy compared to baseline methods. Notably, for methods using ResNet101, HHRM delivers superior results while reducing computational cost, highlighting its advantages.

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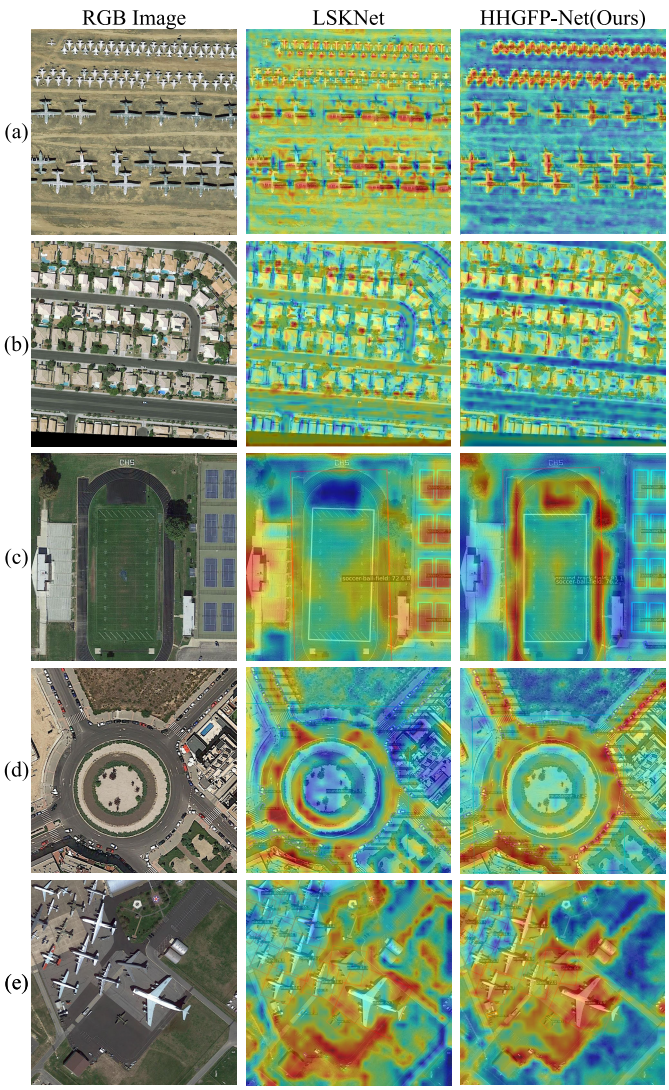

Fig. 9. Qualitative comparison of attention maps in the backbone network. Compared with LSKNet, a method with multiscale feature perception and large receptive fields, HHGFP-Net obtains more discriminative feature representations when detecting objects of different scales. (a)–(e) Several typical scenarios from the DOTA dataset.

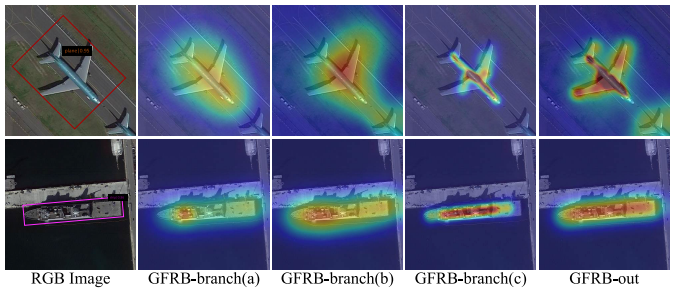

Fig. 10. Qualitative analysis of GFRB. The rich shape information extracted through multiple pathways significantly enhances GFRB's perception and learning of object geometric poses. This improved capability allows for a more accurate modeling of object orientation and shape details in complex remote sensing scenarios.

3) Ablation Experiments on GFP-FPN: GFP-FPN consists of two sub-modules: GFRB, designed to adapt to the geometric shapes of objects in RSIs, and PFEB, which aims to extract consistent information across multilevel features while enhancing prominent foreground features for more accurate representations, especially for small targets. The ablation study results for GFRB and PFEB in Table VIII show that adding GFRB to the baseline method achieved a significant 0.32% improvement in accuracy compared to traditional FPN. Employing PFEB can also improve detection results with a slight increase in additional parameters and FLOPs. By integrating both GFRB and PFEB, GFP-FPN achieves the optimal accuracy improvement, reaching 80.68%. To better illustrate the advantages of GFRB in learning the geometric orientations and shapes of objects, we visualized the features processed by GFRB in Fig. 10. The experimental results demonstrate that GFRB effectively integrates features from multiple pathways and accurately models the geometric shapes of objects, thereby providing precise feature representations. In addition, Fig. 11 presents several qualitative comparisons between GFP-FPN and the conventional FPN. It can be observed that GFP-FPN achieves activation regions that are more closely aligned with the actual shapes of objects compared to traditional FPN. Moreover, in complex scenes with intricate interactions between objects and their backgrounds, GFP-FPN effectively identifies prominent foreground features without introducing noticeable background noise.

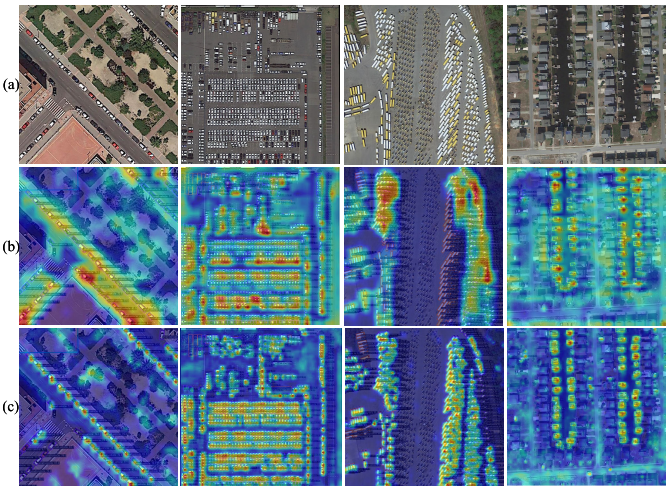

Fig. 11. Qualitative comparison between FPN and the proposed GFP-FPN in several typical scenarios. (a) RGB images. (b) FPN. (c) GFP-FPN.

To validate the advantages of GFP-FPN, we conducted experiments on the STAR dataset, with a specific focus on fine-grained categories such as bridges, ships, and gravity dams. As shown in Table IX, integrating the GFRB improved the mAP of "bridge" by 2.4% and "ship" by 2.5% compared to the conventional FPN. Additionally, the inclusion of the PFEB further enhanced the overall detection accuracy. The comparative analysis results of existing state-of-the-art geometric pose learning methods on the STAR dataset are presented in Table X. HHGFP-Net demonstrates a clear superiority over other methods, particularly in detecting objects such as ships, bridges, and gravity dams that exhibit arbitrary directions and poses. These results align with the qualitative analyses presented in Figs. 10 and 11, which demonstrate that GFP-FPN effectively strengthens the model's ability to learn object shapes and salient features.

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F. Discussion and Limitation Analysis

Our proposed HHGFP-Net surpasses all mainstream methods across four RSOD datasets, achieving state-of-the-art performance. This superiority stems from two key innovations.

1) In multiscale feature extraction, we amplified critical features at various backbone levels while suppressing noise, yielding more discriminative representations.
2) We improved geometric pose learning for remote sensing objects within the neck structure, enhancing foreground feature representation and enabling more precise detection, particularly for smaller objects. However, our approach has three primary limitations. First, the feature compensation across backbone levels relies heavily on empirical experiments, limiting adaptability to diverse datasets. Second, the geometric pose fitting was only tested on a limited range of extreme cases, necessitating further exploration for broader scenarios. At last, the method's high computational complexity caused GPU memory issues when integrated with other algorithms, underscoring the need for lightweight model design in future work.

In the next phase of our research, we will focus on deploying dynamic decision-making and adaptive learning to enhance or suppress features at different backbone levels. Concurrently, we will incorporate shape constraints into the FPN before the final detection stage to improve its capability to learn object poses and foreground features, ultimately achieving better detection performance. These strategies will also be further extended to downstream tasks, such as remote sensing change detection [74] and image dehazing [75].

V. Conclusion

Object detection in aerial images remains a challenging topic due to the large-scale variations and diverse geometric poses of targets in RSIs. In this work, we propose a HHGFP-Net for RSOD. Specifically, the HHRM leverages hierarchical information to guide feature extraction at different levels of the backbone, suppressing interference and enhancing important features, effectively alleviating the limitations in existing multiscale methods, achieving more discriminative feature representation. Furthermore, we propose the GFP-FPN, which aims to learn the geometric poses of remote sensing objects and enhance foreground features by cross-level feature consistency search, thereby providing more precise features for final detection. Experiments on four RSOD datasets demonstrate that our method is superior to other state-of-the-art methods.

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